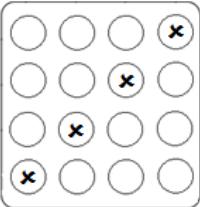


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		award (1) for either of following aluminium is more reactive than iron aluminium displaces iron award (1) for correct reference to reduction e.g. aluminium reduces iron(III) oxide aluminium takes the oxygen away from the iron iron(III) oxide is reduced	2			2		
		(ii)		$\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$ products (1) balancing (1) balancing mark can only be awarded if the correct products given		2		2		
		(iii)		70% (2) if incorrect award (1) for any of following $M_r(\text{Fe}_2\text{O}_3) = 160$ $56 + 56 + 16 + 16 + 16$ total relative mass of iron = 112 $56 + 56$		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		 all four crosses shown (1) award (1) for any of following metals do not displace themselves from solution metals do not react with their own nitrate metals do not react with themselves metals have the same reactivity – neutral answer correct explanation may be credited when the first mark is not awarded	2			2		2
		(ii)		0.381g / 0.38g / 0.4g (3) if answer incorrect credit working $112 \leftrightarrow 190.5 / (2 \times 56) \leftrightarrow (3 \times 63.5) / \frac{0.224}{112}$ (1) $\frac{0.224}{112} \times 190.5$ (1) alternative method 0.004 mol Fe (1) 0.006 mol Cu (1) ecf possible		3		3	3	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		2 $\text{Cl}^- \longrightarrow \text{Cl}_2 +$ 2 e^- (1) award (1) for either of following <u>chloride</u> / Cl^- ions lose electrons oxidation is loss of electrons do not accept - <u>chlorine</u> / zinc chloride loses electrons	2			2		
		(ii)	I	award (1) for any of following some of the zinc chloride/electrolyte is also on the electrode impurities also stick to the electrode product of side reactions stick to the electrode			1	1		1
			II	the chlorine formed is a gas / does not stick to the electrode	1			1		1
				Question 4 total	7	7	1	15	5	4